
Phase 1, Module 4: Operators & Logic Engines

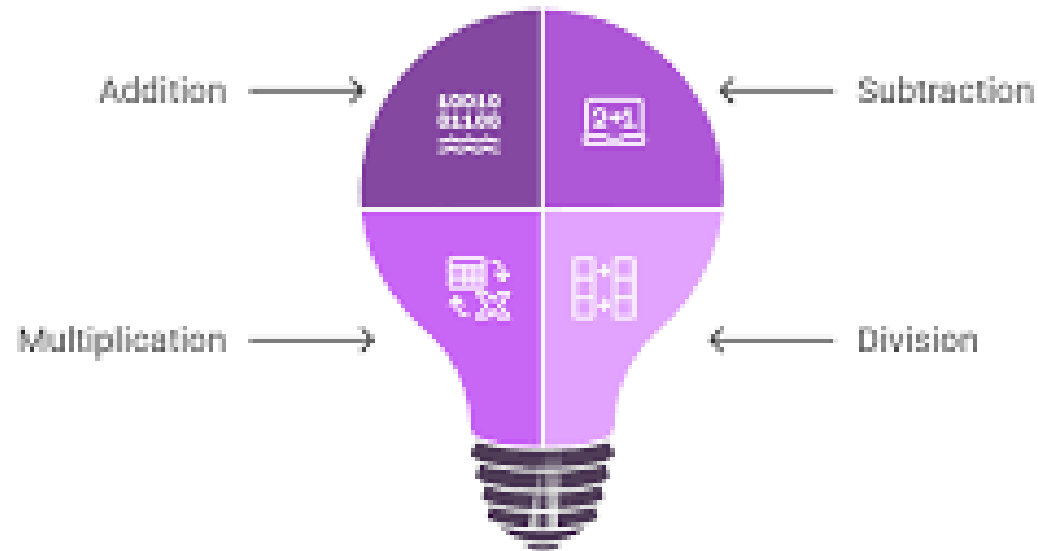
Khyber AI Systems Engineer (KASE)

Welcome to Module 4. An AI is just a massive calculator making logical decisions millions of times per second. Today, we learn the mathematical and logical operators that drive those decisions.

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Core Arithmetic (The AI Lens)

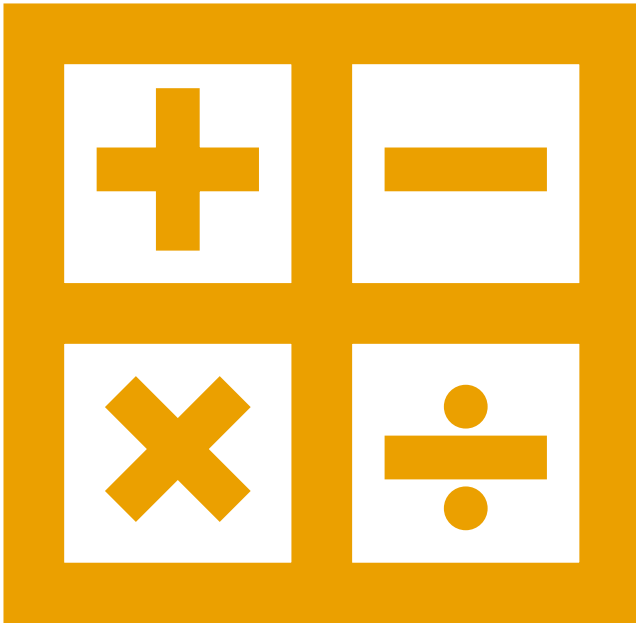
Core Arithmetic Operations in Computing



Feature Scaling (+, -, *, /)

- + / -: Used for adjusting AI "Biases" or centering data.
- *: Used for applying "Weights" to data.
- /: Crucial for Normalization (e.g., dividing image pixels by 255 so the AI can process them faster).

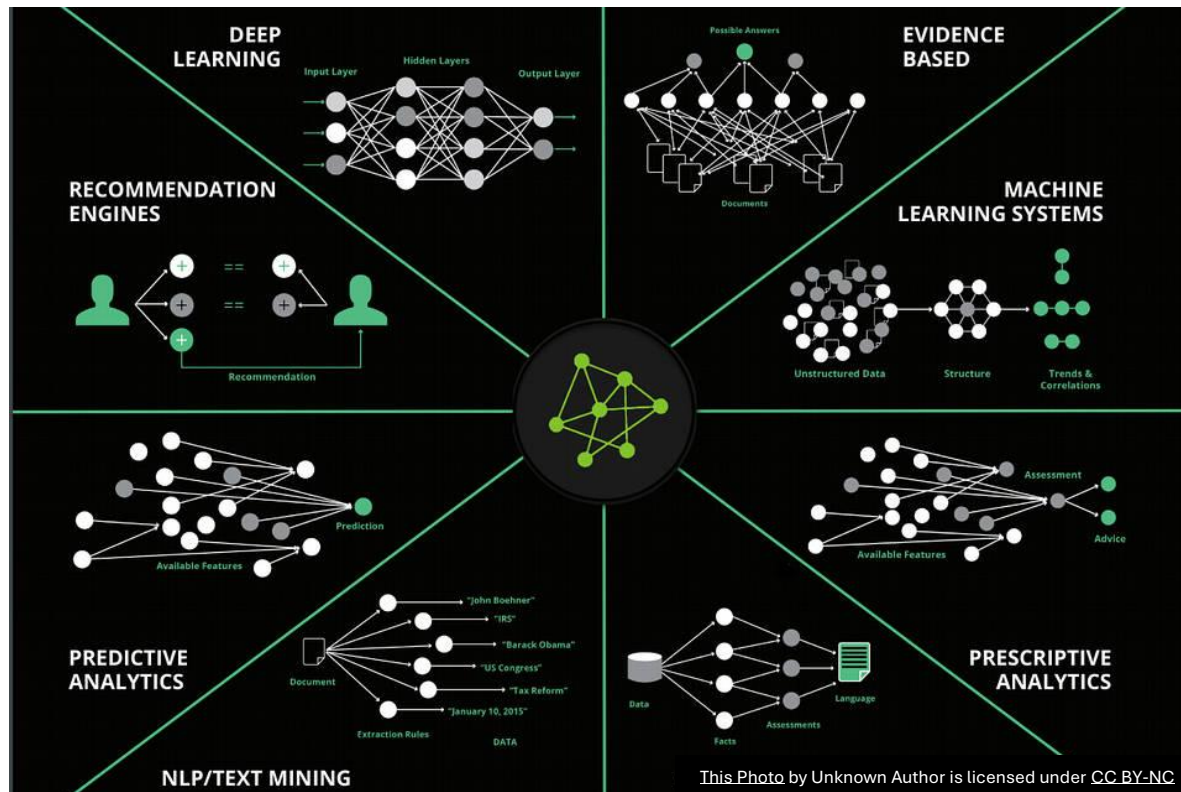
Exponents & The Cost of Being Wrong



Exponentiation (**)

- In Python, `^` is NOT an exponent (it's a bitwise operator).
- Use `**` for powers (e.g., $2^{**}3 = 8$).
- AI Use Case: Used constantly to calculate "Mean Squared Error" — mathematically punishing the AI when it makes a wrong prediction.

The Math of AI (Beyond Addition)



Floor Division & Modulo

- Standard Division ($/$): Always returns a Float (decimal).
- Floor Division ($//$): Chops off the decimal. Returns an Integer. (Crucial for splitting data into batches).
- Modulo ($\%$): Returns the remainder of division. (Used to trigger events, like saving an AI model every 10 epochs).

The Interview Trap (`==` vs `is`)

Value Equality vs. Memory Identity

- `==`: Do these two variables hold the exact same value?
- `is`: Are these two variables pointing to the exact same address in RAM?

```
>>> a = [1, 2, 3]
>>> b = a
>>> c = a[:]

>>> a == b
True
>>> a == c
True

>>> a is b
True
>>> a is c
False
```